Check your understanding

1. Look at your graphs and describe the relationship between current and potential difference for a resistor at a constant temperature. [1 mark]

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2. Explain why the power supply should be turned off between readings. [2 marks]

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3. Calculate the resistance of your resistor at a constant temperature. [2 marks]

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4. Describe how you can improve the accuracy of this calculated result. [2 marks]

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Exam-style questions

1. When the potential difference across a filament lamp is 6.6 V, the current is 0.3 A.

   Calculate the resistance. Give the correct unit. [2 marks]

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2. A student wants to investigate how the current through a diode affects its resistance.

   Describe how the student could use a circuit to investigate how current through a diode affects the resistance of the diode. [6 marks]

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4.2.1.4 I–V characteristics

Check your understanding

1. As potential difference increases, current increases (in both directions). [1]
2. So that the temperature of the resistor is kept constant. [1]
   Resistance increases as temperature increases. [1]
3. Use any pair of values from results table. [1]
   Calculate using $R = V/I$ [1]
4. Repeat each reading. [1]
   Calculate a mean. [1]

Exam-style questions

1. correct substitution into formula: 6.6/0.3 [1]
   resistance = 22 Ω [1]
2. Answer should include: [6]
   - Ammeter used to measure current.
   - Voltmeter used to measure potential difference.
   - Resistance of variable resistor altered to change current in circuit or change potential difference (across diode).
   - Plot a graph of current against potential difference.
   - Resistance (of diode) calculated from pairs of points on the graph, for current in both directions using R = V/
   - Resistance calculated for a large enough range of currents, flowing in both directions.